

AI-POWERED PLATFORM FOR TALENT DISCOVERY AND SHOWCASE

BSc. in Software Engineering

Isaac Mugisha

Supervisor: [Supervisor Name]

African Leadership University

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ABSTRACT

Talented creators in East Africa, particularly those operating in Rwanda's emerging creative economy, face a persistent structural problem: their work remains invisible on major digital platforms despite its quality. Existing discovery platforms rely on engagement-driven recommendation algorithms that systematically favour creators who already have large audiences, compounding early disadvantage and producing outcomes that are both commercially inefficient and socially inequitable. This research proposes the design, development, and evaluation of an AI-powered talent discovery and showcase platform that addresses this gap through machine learning-based content quality assessment, fairness-constrained recommendation, and personalised audience matching. The platform targets emerging musicians, visual artists, comedians, athletes, and performing artists aged 18 to 45 operating in Kigali, Rwanda, as the primary pilot population. A mixed-methods research design will be employed, combining quantitative platform performance evaluation with qualitative user experience data gathered through surveys, semi-structured interviews, and focus group discussions. A pilot involving approximately 100 to 150 creator accounts and 300 to 500 audience accounts will be conducted over eight to twelve weeks. The recommendation algorithm will be evaluated against standard information retrieval metrics including precision at k and recall at k , as well as fairness metrics measuring demographic parity across gender and geographic subgroups. It is hypothesised that a fairness-constrained recommendation system will increase visibility for emerging creators by at least 30 percent compared to engagement-only baselines while maintaining audience engagement rates within 10 percent of those baselines. The findings are expected to produce a deployable platform prototype, a validated methodology for equitable talent discovery, and an empirical contribution to the limited body of research addressing recommendation systems in sub-Saharan African creative economy contexts.

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CHAPTER ONE: INTRODUCTION

1.1 Introduction and Background

The global creative economy is one of the fastest-growing sectors of the digital age. According to the United Nations Conference on Trade and Development, creative industries generated over USD 2.25 trillion in output in 2022 and employed more than 50 million people worldwide, with young people aged 15 to 29 disproportionately represented among active creators (UNCTAD, 2022). In sub-Saharan Africa, this growth is particularly pronounced. Rwanda's creative sector has expanded significantly over the last decade, driven by increasing smartphone penetration, a growing urban middle class, and government-backed digital economy initiatives under Vision 2050. In Kigali alone, the number of registered creative professionals — including musicians, visual artists, comedians, and performing artists — increased by an estimated 40 percent between 2018 and 2023 (Rwanda Development Board, 2023). Despite this growth, the majority of these creators remain economically marginalised and professionally invisible beyond their immediate social networks.

Traditionally, talent discovery in the Rwandan creative economy has depended on informal word-of-mouth networks, in-person showcase events, and relationships with a small number of local talent agents and event promoters. These channels are not scalable, are heavily influenced by social connections rather than creative merit, and systematically exclude creators who lack the capital or social capital to access them. The arrival of global digital platforms such as YouTube, Instagram, TikTok, and Spotify appeared to offer an alternative, providing creators with direct access to global audiences without intermediaries. However, the algorithmic architecture of these platforms has reproduced, and in some cases amplified, the structural inequities that existed in offline talent discovery. Research consistently demonstrates that recommendation algorithms on major platforms allocate disproportionate visibility to content that already has high engagement signals — views, likes, shares, and follower counts — creating a Matthew effect in which early advantage compounds over time and new or emerging creators struggle to gain initial traction regardless of the quality of their work (Celma, 2010; Ferraro et al., 2021).

The gap between existing platforms and the needs of East African emerging creators motivates this research. This project proposes the design, development, and evaluation of an AI-powered

talent discovery platform specifically engineered to address the structural barriers that prevent quality creative work from achieving visibility proportionate to its merit. The platform integrates machine learning-based content quality assessment, fairness-constrained recommendation algorithms, and personalised audience matching to create a more equitable talent discovery ecosystem. The pilot deployment focuses on Kigali, Rwanda, where the researcher has direct contextual familiarity with the creative community and where the gap between creator supply and audience reach is well documented.

1.2 Problem Statement

The first sub-problem is the structural invisibility of quality creative work on existing digital platforms. Major content discovery platforms rely on engagement metrics — view counts, follower numbers, and interaction rates — as the primary signals for content promotion. This architecture creates a feedback loop in which content from already-popular creators receives greater algorithmic amplification, while equally competent or superior work from emerging creators receives negligible exposure. Ferraro et al. (2021) demonstrated that this engagement-first model leads to systematic underrepresentation of new entrants across music streaming platforms, with the bottom 80 percent of creators by follower count receiving less than 5 percent of total recommendation-driven streams. In the East African context, this problem is compounded by the fact that local creators are starting from an even lower baseline of followers and engagement signals compared to creators in markets where these platforms were originally designed and optimised (Nwagwu & Akintoye, 2024).

The second sub-problem is the absence of content quality assessment mechanisms that are independent of popularity signals. Current platforms have no reliable, scalable method for evaluating the technical or artistic quality of creative content in a way that is decoupled from existing engagement data. Human curation at scale is economically impractical, and existing automated content moderation systems are designed for compliance rather than quality assessment. Without an independent quality signal, platform operators have no basis on which to promote emerging creators ahead of their engagement metrics, meaning that quality-merit-based discovery is structurally unavailable as a mechanism for creator advancement. This gap has been identified as one of the most significant unresolved challenges in recommendation systems research for creative domains (Schedl et al., 2018; Deldjoo et al., 2021).

The third sub-problem is the mismatch between available audience matching tools and the specific characteristics of East African creative content and audiences. Most recommendation systems are trained predominantly on datasets from North American and European markets, embedding cultural assumptions about content quality, genre boundaries, and audience preferences that do not translate to the Rwandan or broader East African context. The result is that even when East African creators do achieve some presence on global platforms, their content is systematically misclassified or under-recommended to audiences who would be most likely to engage with it. This data representation gap has been documented by Birhane et al. (2021) as a systemic problem across AI systems built on globally unrepresentative training data. This work introduces an AI-powered platform that integrates a culturally aware content classification model, a fairness-constrained recommendation engine, and a merit-based visibility scoring system, which this research aims to design, build, and empirically evaluate in order to address the three problems described above.

1.3 Project's Main Objective

The overall aim of this project is to develop and evaluate an AI-powered software platform that enables equitable talent discovery and showcase for emerging creative professionals in Rwanda by providing merit-based content quality assessment and fairness-constrained recommendation, thereby addressing the structural visibility gaps documented in the problem statement.

1.3.1 Specific Objectives

1. To review existing literature on recommendation systems, algorithmic fairness, and creative economy platforms, and to collect baseline data on the discovery challenges facing emerging creators in Kigali, Rwanda.
2. To design and develop an AI-powered platform incorporating a machine learning-based content quality assessment model, a fairness-constrained collaborative filtering recommendation engine, and a creator profile and showcase interface.
3. To evaluate the platform against measurable performance metrics including content categorisation accuracy (target: 85% or above), recommendation precision and recall at k, and creator visibility improvement rates, as well as fairness metrics assessing demographic parity across gender and geographic subgroups.

1.4 Research Questions

4. To what extent does a fairness-constrained recommendation algorithm improve visibility outcomes for emerging creators in Kigali, Rwanda compared to an engagement-only baseline recommendation system?
5. How accurately can a machine learning model assess the technical quality of creative content across diverse disciplines — including music, visual art, comedy, and athletics — without relying on engagement signals?
6. How do creators and audiences in Kigali, Rwanda experience and evaluate the fairness, usability, and utility of an AI-powered talent discovery platform compared to existing platforms they currently use?

1.5 Project Scope

Geographically, this research focuses on Kigali, Rwanda as the primary pilot environment. Kigali has been selected because it concentrates the majority of Rwanda's formal creative sector, offers growing digital infrastructure, and provides the researcher with direct contextual access to relevant participant communities. The platform will be tested with a sample of creators and audience members drawn from Kigali's creative community rather than attempting national or pan-African reach in the pilot phase. National and regional scale-up is a post-pilot objective.

In terms of creative disciplines, the platform will cover five categories in the pilot: music, visual arts, comedy, athletics, and performing arts. These categories reflect the dominant creative disciplines in Kigali's formal and informal creative sector and align with the platform's content classification model's initial training scope.

Technically, the platform scope is limited to a web-based application with mobile browser compatibility. A dedicated native mobile application is out of scope for the pilot phase. The recommendation engine will focus on collaborative and content-based filtering; reinforcement learning-based recommendation improvements are planned for a later development phase. The pilot will run for eight to twelve weeks, with a minimum of 100 creator accounts and 300 audience accounts required to generate meaningful recommendation data.

1.6 Significance and Justification

Upon successful implementation, this platform has the potential to measurably increase the economic opportunities available to emerging creative professionals in Kigali. Research on the economic impact of improved online visibility for small and independent creators documents income gains of between 25 and 60 percent for creators who achieve consistent algorithmic promotion on major platforms (McKinsey & Company, 2021). By extending this kind of visibility to creators who are currently excluded by engagement-first algorithms, the platform could contribute directly to Rwanda's creative economy growth targets under Vision 2050.

The project also makes a methodological contribution to the field of AI-powered recommendation systems. There is currently very limited empirical research on recommendation system performance in low-data, culturally specific environments in sub-Saharan Africa. The evaluation methodology developed in this project, including culturally adapted content quality metrics and fairness evaluation frameworks, will provide a replicable template for similar research across the East African region and beyond.

Finally, this research contributes to the growing body of work on algorithmic fairness in AI systems. By designing, implementing, and empirically testing a fairness-constrained recommendation engine in a real-world deployment context, it provides concrete evidence on the practical trade-offs between recommendation accuracy and equity — a question that has been extensively theorised but rarely tested at deployment scale in African contexts.

1.7 Research Budget

Item	Estimated Cost (USD)	Notes
Cloud hosting (AWS/DigitalOcean, 3 months)	\$90	Basic compute for pilot
Domain name registration (1 year)	\$15	Platform domain
Data labelling (content annotation)	\$120	Human annotation for model training

Participant transport and refreshments	\$80	Focus groups and interviews
Translation support (Kinyarwanda)	\$50	Interview transcription
Total	\$355	

1.8 Research Timeline

Activity	Week 1-2	Week 3-4	Week 5-6	Week 7-8	Week 9-10	Week 11-12	Week 13-14
Platform development and testing	✓	✓	✓				
Ethical approval process	✓	✓					
Participant recruitment			✓	✓			
Pilot deployment (platform live)				✓	✓	✓	
Surveys (baseline, midpoint, post)				✓	✓	✓	
Interviews and focus groups					✓	✓	
Data analysis and write-up						✓	✓

CHAPTER TWO: LITERATURE REVIEW

2.1 Introduction

This chapter reviews existing literature relevant to three interconnected domains: the creative economy and talent discovery, digital platform recommendation systems, and algorithmic fairness in AI. The search was conducted across four indexed academic databases — Google Scholar, IEEE Xplore, ACM Digital Library, and Semantic Scholar — using the search terms 'talent discovery', 'music recommendation systems', 'algorithmic fairness in recommender systems', 'creative economy Africa', and 'content-based filtering creative content'. The review focused on software-related systems and empirical studies published between 2015 and 2025, with some foundational theoretical works included from earlier periods. Approximately 60 papers and reports were identified in the initial search, of which 32 were selected for detailed review based on their direct relevance to the research questions. The review is organised thematically around existing systems, related technical work, and identified strengths and weaknesses.

2.2 Overview of Existing Systems

Several software platforms currently address aspects of the talent discovery problem, though none does so in a manner that fully addresses the needs of emerging creators in the East African context.

Spotify's Discover Weekly and Release Radar features use collaborative filtering augmented by audio feature analysis to recommend music to existing listeners. While technically sophisticated, Spotify's system is explicitly designed to optimise engagement metrics for its existing user base, which is predominantly located in North America and Europe (Schedl et al., 2018). The cold-start problem — the difficulty of generating accurate recommendations for new creators with limited listening data — has been documented as a persistent limitation that disproportionately affects new and independent artists (Celma, 2010).

TikTok's For You Page algorithm represents the current state of the art in short-form video recommendation and has been widely credited with enabling some degree of viral discovery for previously unknown creators. However, research indicates that TikTok's algorithm still exhibits strong engagement compounding behaviour: creators who achieve high engagement in their first few posts receive substantially more algorithmic promotion than those whose early content

performs modestly, regardless of subsequent quality improvements (Ferraro et al., 2021). Furthermore, TikTok's training data and content moderation standards have been documented to embed cultural biases against African and Global South content creators (Birhane et al., 2021).

YouTube's recommendation system similarly relies heavily on watch time and click-through rate as its primary recommendation signals. Studies of YouTube's recommendation behaviour document a strong rich-get-richer dynamic in which the top 3 percent of channels by subscriber count receive over 85 percent of algorithmically recommended views (Ribeiro et al., 2020). The platform's Partner Program, which provides revenue sharing to creators, requires a minimum of 1,000 subscribers and 4,000 watch hours, creating a threshold that many emerging East African creators cannot reach without initial algorithmic support.

In the East African context specifically, platforms such as Mdundo (Kenya) and Boomplay (pan-African) have attempted to build locally relevant music streaming services. However, both platforms rely primarily on curator-driven playlist promotion rather than algorithmic recommendation, are limited to the music discipline, and do not offer content quality assessment or creator development tools. Neither platform has published empirical research on the equity or effectiveness of its discovery mechanisms.

2.3 Review of Related Work

2.3.1 Recommendation Systems in Creative Domains

Recommender systems in creative domains have been studied extensively over the past two decades, with music recommendation receiving the most academic attention. Schedl et al. (2018) provide a comprehensive survey of music recommendation approaches, identifying three primary paradigms: collaborative filtering (which recommends content based on the listening behaviour of similar users), content-based filtering (which recommends content based on audio or visual features of the content itself), and hybrid approaches combining both. The survey finds that hybrid approaches consistently outperform either method in isolation on standard benchmarks but that both approaches suffer from the popularity bias problem: because training data is drawn from existing platform usage, popular content is over-represented and emerging content is systematically under-recommended.

Deldjoo et al. (2021) specifically address the problem of fairness in recommendation systems, reviewing 29 papers on the topic and identifying two primary framings: consumer fairness (ensuring that users from different demographic groups receive equally good recommendations) and producer fairness (ensuring that creators from different demographic groups receive equitable exposure). The review finds that producer fairness has received far less research attention than consumer fairness, and that most existing fairness interventions focus on post-processing the output of existing recommendation algorithms rather than building fairness constraints into the model architecture. This finding directly motivates the approach taken in the current project.

Ferraro et al. (2021) conduct an empirical analysis of popularity bias in music recommendation using data from a major streaming platform, finding that the top 20 percent of artists by popularity receive more than 80 percent of recommended listening sessions. Their work demonstrates that applying a re-ranking step with a diversity penalty significantly reduces this concentration without substantially reducing user satisfaction, supporting the feasibility of the fairness-constrained recommendation approach proposed in this research.

2.3.2 Content Quality Assessment Using Machine Learning

Automated content quality assessment is a less mature research area than recommendation, but a growing body of work demonstrates the feasibility of using machine learning to assess technical quality in creative domains. In music, models trained on audio features including spectral centroid, mel-frequency cepstral coefficients, and rhythmic regularity have achieved accuracy rates above 80 percent on quality classification tasks when trained with sufficient labelled data (Müller, 2015). Liu et al. (2018) demonstrate that convolutional neural networks applied to spectrograms can distinguish between professionally and amateurly produced music with an accuracy of 87 percent, even in the absence of metadata signals.

For visual content, pre-trained convolutional neural network models including ResNet and VGG have been adapted for aesthetic quality assessment tasks with strong results, with Talebi and Milanfar (2018) reporting a correlation of 0.73 between model predictions and human aesthetic ratings on the AADB dataset. These results suggest that content quality assessment models can provide meaningful quality signals that are independent of engagement data, which is a core requirement for the platform proposed in this research.

2.3.3 Algorithmic Fairness Frameworks

The theoretical basis for the fairness constraints implemented in the platform recommendation engine draws on two primary bodies of work. Binns (2017) provides a philosophical analysis of fairness in machine learning, arguing that different fairness criteria — including individual fairness, group fairness, counterfactual fairness, and calibration — are often mutually incompatible and that the choice among them is fundamentally a value judgment rather than a technical question. For this research, demographic parity (ensuring that the recommendation system allocates exposure proportionally across gender and geographic subgroups of creators) and equalized opportunity (ensuring that high-quality creators from all demographic groups have an equal probability of being recommended) are adopted as the primary fairness criteria, consistent with the producer-side fairness framing advocated by Deldjoo et al. (2021).

Jobin et al. (2019) conduct a global meta-analysis of AI ethics guidelines, identifying transparency, accountability, and fairness as the three most consistently cited principles across 84 governance frameworks from 37 countries. Their finding that transparency — providing users with intelligible explanations of algorithmic decisions — is cited in 73 of 84 frameworks reviewed informs the platform design decision to include plain-language algorithmic explanation features in the creator dashboard.

2.3.4 The East African Creative Economy and Digital Platforms

UNCTAD (2022, 2024) provide the most comprehensive available data on the African creative economy, documenting both its rapid growth and its structural underdevelopment relative to its potential. The 2024 report specifically identifies digital platform access and algorithmic visibility as key barriers to income generation for African creators, recommending investment in locally appropriate platform infrastructure as a priority for creative economy development policy.

Nwagwu and Akintoye (2024) examine the relationship between social media platform behaviour and music career development for emerging African artists, finding that the algorithmic design of global platforms creates systematic barriers for African creators even when those creators achieve initial audience traction. Their study of 150 emerging Nigerian musicians documents that over 70 percent reported experiencing algorithm-related visibility drops after initial promotional periods,

a pattern consistent with the engagement compounding behaviour documented in the broader recommendation systems literature.

2.3.5 Summary of Reviewed Literature

Author(s) & Year	Topic	Key Finding	Relevance to This Study
Schedl et al. (2018)	Music recommendation survey	Hybrid CF+CBF outperforms either alone; popularity bias is persistent	Justifies hybrid recommendation approach
Deldjoo et al. (2021)	Fairness in recommender systems	Producer fairness under-researched; post-processing is dominant but inadequate	Motivates in-model fairness constraints
Ferraro et al. (2021)	Popularity bias in music streaming	Re-ranking with diversity penalty reduces bias without harming UX	Supports feasibility of fairness approach
Binns (2017)	Philosophical analysis of ML fairness	Fairness criteria are value-laden and often mutually incompatible	Grounds choice of fairness metrics
Jobin et al. (2019)	Global AI ethics guidelines	Transparency and accountability cited in 73/84 frameworks reviewed	Informs platform transparency design
UNCTAD (2022, 2024)	African creative economy	Platform visibility is key barrier for African creators	Establishes real-world problem context
Nwagwu & Akintoye (2024)	Social media and African music careers	Over 70% of emerging artists report algorithm-related visibility drops	Direct empirical evidence of the problem
Birhane et al. (2021)	Bias in AI training datasets	Global AI systems embed systematic bias against African content	Motivates locally trained classification model

2.4 Strengths and Weaknesses of Existing Systems

The existing platforms reviewed offer several genuine strengths. Spotify's hybrid recommendation architecture achieves strong personalisation performance on its target user base and has invested significantly in audio feature analysis, producing content classification models that are technically sophisticated. TikTok's For You Page algorithm is demonstrably effective at surfacing novel content to engaged users and has enabled some degree of cross-cultural discovery that was not possible on previous platforms. YouTube's recommendation system benefits from the world's largest video training dataset, giving it unparalleled ability to match content to existing viewer preferences.

However, all existing platforms share critical weaknesses from the perspective of equitable talent discovery. First, all systems are built on engagement-driven feedback loops that systematically disadvantage creators without existing audiences. Second, none of the major platforms offers a quality assessment mechanism that is independent of engagement signals, meaning that technical quality cannot serve as an independent promotion pathway. Third, all major platforms are trained primarily on data from Global North contexts and embed cultural assumptions that do not translate to East African creative content and audiences. Fourth, none of the platforms reviewed offers transparent, creator-facing explanations of how algorithmic decisions affecting their visibility are made, leaving creators without the information needed to develop effective platform strategies. The locally oriented platforms (Mdundo, Boomplay) address some cultural relevance concerns but lack any algorithmic sophistication in their discovery mechanisms, rely on manual curation, and are limited to the music discipline.

2.5 General Comment and Conclusion

The literature review reveals a clear and well-documented gap between what existing talent discovery platforms offer and what emerging creators in East Africa need. The theoretical and empirical foundations for addressing this gap through fairness-constrained recommendation, machine learning-based content quality assessment, and mixed-methods evaluation are well established in the literature. The specific combination of these approaches, applied to the East African creative economy context with a locally relevant training dataset and pilot deployment,

represents a genuine contribution to both the research literature and the practical platform landscape. The following chapter describes the system design that operationalises these insights.

CHAPTER THREE: SYSTEM ANALYSIS AND DESIGN

3.1 Introduction

This chapter describes the technical design of the AI-powered talent discovery platform. The research methodology guiding the platform's evaluation is a mixed-methods design, combining quantitative performance assessment with qualitative user experience research, as detailed in the Unit Four methodology chapter. The system design translates the research objectives and literature-informed requirements into a concrete software architecture. The design follows a modular, layered approach in which the frontend presentation layer, the API business logic layer, the machine learning inference layer, and the data storage layer are loosely coupled to allow independent development, testing, and scaling. All architectural decisions are justified by reference to the requirements established in the problem statement and the gaps identified in the literature review.

3.2 Research Design and Development Model

The development methodology adopted for this project is an adapted Agile iterative model, modified to accommodate the research evaluation requirements of a mixed-methods academic study. The development process is organised into four phases: requirements and design (weeks one to four), core platform development (weeks five to eight), pilot deployment and data collection (weeks nine to fourteen), and analysis and reporting (weeks fifteen to sixteen). This phased approach ensures that the platform is sufficiently stable for research deployment while allowing iterative refinement based on user feedback collected during the pilot.

The machine learning pipeline follows a standard MLOps model. A pre-training phase uses publicly available audio and image datasets to build initial content quality and classification models. A fine-tuning phase uses pilot-generated platform data to adapt these models to the specific characteristics of Rwandan creative content. A re-ranking fairness layer is applied at inference time to enforce demographic parity constraints without requiring model retraining. This design is informed by the post-processing fairness intervention approach reviewed in Deldjoo et al. (2021), adapted to operate closer to the model architecture level than typical post-processing approaches.

3.3 System Architecture

The platform is built on a modular, layered architecture ensuring that data flows efficiently from user input through processing to output. The primary layers are as follows.

The frontend layer is a React.js web application providing the creator dashboard, content upload interface, audience discovery interface, and platform analytics visualisations. The frontend is optimised for mobile browser access given the smartphone-dominant internet access patterns of the target user population in Kigali. The creator dashboard includes a plain-language algorithmic transparency panel showing creators how their visibility score is calculated, consistent with the transparency principle identified by Jobin et al. (2019).

The backend layer is a Spring Boot REST API handling authentication, content ingestion, recommendation serving, and analytics aggregation. Spring Boot is selected for its mature ecosystem support for microservices architecture, which allows the recommendation engine, content quality assessment service, and user management service to be deployed and scaled independently. JWT-based authentication is implemented to secure all API endpoints. The backend connects to a PostgreSQL relational database for structured data — user profiles, content metadata, recommendation logs — and to an Amazon S3-compatible object store for audio and video file storage.

The machine learning layer is a Python-based FastAPI service that exposes endpoints for content quality scoring and recommendation generation. Content quality models are built using PyTorch, with pre-trained ResNet-50 and VGG-16 architectures fine-tuned for visual content quality assessment and a mel-spectrogram convolutional model used for music quality scoring. The recommendation engine uses a matrix factorisation collaborative filtering model augmented with content-based features and a fairness re-ranking layer that enforces demographic parity constraints at serving time. The ML service is containerised using Docker to ensure environment consistency across development and production deployments.

3.4 Machine Learning Pipeline

3.4.1 Content Quality Assessment

The content quality assessment component evaluates uploaded creative content across five dimensions: technical production quality, structural coherence, originality indicators, genre-appropriateness, and audience accessibility. For music content, audio features including mel-frequency cepstral coefficients, spectral centroid, zero-crossing rate, and rhythmic regularity are extracted using the Librosa library and passed through a two-layer convolutional neural network trained on the FMA (Free Music Archive) dataset and fine-tuned on locally collected Rwandan music samples. For visual arts, a ResNet-50 model pre-trained on ImageNet is fine-tuned on the AADB aesthetic quality dataset supplemented by locally annotated visual content samples. Quality scores are normalised to a 0 to 100 scale and incorporated into the creator visibility score alongside engagement data.

The project will benchmark three model architectures for content quality classification: a baseline random forest on hand-crafted features, a convolutional neural network on raw audio/image features, and a pre-trained transformer model. Selection of the final architecture will be based on classification accuracy, inference latency, and calibration of predictions across creator demographic subgroups.

3.4.2 Recommendation Engine

The recommendation engine uses a hybrid collaborative and content-based filtering approach. The collaborative filtering component is implemented using matrix factorisation with alternating least squares, trained on the user-content interaction matrix generated from platform usage data. The content-based component generates item embeddings from the quality assessment models and creator metadata. At serving time, candidate content items are scored by combining collaborative and content-based scores, and a fairness re-ranking step is applied that adjusts scores to ensure that the distribution of recommended content across gender and geographic subgroups of creators reflects demographic parity. The re-ranking approach follows the proportional fairness intervention proposed by Biega et al. (2018), which minimises the deviation from merit-based ranking while enforcing the demographic parity constraint.

3.5 UML Diagrams

This section presents the primary UML diagrams defining the system structure and behaviour. The following diagrams are included: use case diagram, class diagram, entity relationship diagram (ERD), and sequence diagram.

3.5.1 Use Case Diagram

The primary actors are Creator, Audience Member, and Platform Administrator. Key use cases include: Upload Content (Creator), View Visibility Dashboard (Creator), Discover Content (Audience Member), Follow Creator (Audience Member), Generate Recommendations (System), Assess Content Quality (System), and Manage User Accounts (Administrator). The Creator and Audience Member actors interact with the system through the React.js web interface. The System actor represents automated processes triggered by user actions.

3.5.2 Class Diagram

Primary classes include User (with subclasses Creator and AudienceMember), ContentItem (with subclasses AudioContent, VideoContent, and ImageContent), RecommendationEngine, QualityAssessmentModel, VisibilityScore, and AnalyticsDashboard. The Creator class has a one-to-many relationship with ContentItem. The RecommendationEngine class has dependency relationships with QualityAssessmentModel and VisibilityScore. The AnalyticsDashboard class has an association with Creator and aggregates VisibilityScore objects.

3.5.3 Entity Relationship Diagram

The primary entities are USER, CONTENT_ITEM, RECOMMENDATION, QUALITY_SCORE, and INTERACTION. USER has a one-to-many relationship with CONTENT_ITEM (a creator can upload many items). CONTENT_ITEM has a one-to-one relationship with QUALITY_SCORE. RECOMMENDATION has a many-to-one relationship with CONTENT_ITEM. INTERACTION (recording audience engagement events) has a many-to-one relationship with both USER and CONTENT_ITEM, with interaction type, timestamp, and session duration as relationship attributes.

3.5.4 Sequence Diagram

The primary sequence documents the content recommendation flow: (1) AudienceMember sends a discovery request to the React.js Frontend; (2) Frontend sends an authenticated request to the

Spring Boot API Gateway; (3) API Gateway calls the ML Service recommendation endpoint with user ID; (4) ML Service retrieves user interaction history from the PostgreSQL Database; (5) ML Service retrieves content quality scores from the Quality Score cache; (6) ML Service generates candidate recommendations using the hybrid model; (7) ML Service applies the fairness re-ranking layer; (8) ML Service returns ranked recommendation list to API Gateway; (9) API Gateway fetches content metadata from PostgreSQL; (10) API Gateway returns formatted recommendation response to Frontend; (11) Frontend renders the discovery page for the AudienceMember.

3.6 Development Tools

Tool / Technology	Category	Justification
React.js	Frontend	Component-based UI library with strong mobile browser support and ecosystem maturity
Spring Boot	Backend API	Researcher's existing production experience; strong support for microservices and JWT auth
PostgreSQL	Relational database	ACID compliance, JSON support for flexible content metadata
FastAPI (Python)	ML service API	High performance async Python API; strong integration with PyTorch
PyTorch	ML framework	Leading framework for CNN and transformer model development
Librosa	Audio feature extraction	Industry standard library for music information retrieval
Docker	Containerisation	Ensures environment consistency across development and deployment
GitHub Actions	CI/CD	Automated testing and deployment pipeline
AWS S3-compatible storage	Media storage	Scalable object storage for audio, video, and image content

Jupyter Notebook	ML experimentation	Model training, hyperparameter tuning, and evaluation
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